Convex Analysis and Monotone Operator Theory in Hilbert Spaces

Chapter 7: Support Functions and Polar Sets

Based on Bauschke & Combettes

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Chapter Overview

Outline

Notation You'll Need I

The Ambient Space

Throughout, $\mathcal{H}$ is a real **Hilbert space** with inner product $\langle \cdot \mid \cdot \rangle$ and norm $\| \cdot \|$. All figures and numeric examples in this deck take $\mathcal{H} = \mathbb{R}^2$ (or $\mathbb{R}$), so you can picture every set as a region in the plane and every $u \in \mathcal{H}$ as an arrow from the origin.

Notation You'll Need II

Carried Over from Earlier Chapters

- $C \subset \mathcal{H}$: an arbitrary subset (often closed and convex).
- $\operatorname{conv} C$: the **convex hull** of C (smallest convex set containing C); $\overline{\operatorname{conv}}\, C$ is its closure, the **closed convex hull**.
- $\operatorname{bdry} C$, $\operatorname{int} C$: the boundary and interior of C.
- $N_C x$: the **normal cone** to C at x (Chapter 6), $N_C x = \{u \in \mathcal{H} : \sup\langle C - x \mid u\rangle \le 0\}$ for $x \in C$, and $N_C x = \varnothing$ for $x \notin C$.
- P_C: the **metric projection** onto a closed convex C.
- $K^{\ominus}$: the **polar cone** of a cone K (Chapter 6), $K^{\ominus} = \{u \in \mathcal{H} : \sup\langle K \mid u\rangle \le 0\}$; $V^{\perp}$: the **orthogonal complement** of a linear subspace V.
- $\operatorname{lev}_{\le 1} f = \{u : f(u) \le 1\}$: the sublevel set of a function f at height 1.

Notation You'll Need III

New in This Chapter

- $\operatorname{spts} C$: the set of **support points** of C (Section 7.1).
- $\sigma_C : \mathcal{H} \to [-\infty, +\infty]$: the **support function** of C (Section 7.2), $\sigma_C(u) = \sup\langle C \mid u\rangle$.
- $C^{\odot}$: the **polar set** of C (Section 7.3), $C^{\odot} = \{u \in \mathcal{H} : (\forall x \in C)\ \langle x \mid u\rangle \leq 1\}$.
- $\|\cdot\|_*$: the **dual norm** of a norm $\|\cdot\|$ on a finite-dimensional space.
- $\ell^2(\mathbb{N})$, $\ell^2_+(\mathbb{N})$: square-summable real sequences, and those with all nonnegative terms (a closed convex cone with empty interior in $\ell^2(\mathbb{N})$).

Notation You'll Need IV

The Running Examples (used throughout this deck)

Two shapes anchor every computation below:

$$C_{\text{disk}} = \{x \in \mathbb{R}^2 : \|x\| \le 1\} \quad \text{(closed unit disk)}, \qquad C_{\text{sq}} = [-1,1]^2 \quad \text{(closed unit square)}.$$

We repeatedly compute $\sigma_C(u)$ and $C^{\odot}$ for these two sets, so the abstract definitions always have concrete numbers attached.

Why This Chapter Matters

The Big Picture

A closed convex set C can be described in two dual ways: as a *collection of points*, or as an *intersection of half-spaces* that pin it down from the outside. The **support function** is the bridge between these two pictures: it packages, for every direction u, the single number "how far can I go into C if I walk in direction u?" Knowing $\sigma_C(u)$ for *every* direction u turns out to completely determine C (when C is closed and convex) — this is the content of Corollary 7.12 below.

Plain-English Preview

- **Support point**: a point of C that sits on some supporting hyperplane — a point you can "touch" from outside with a flat wall that leaves all of C on one side.
- **Support function** $\sigma_C(u)$: the maximum value of $\langle x \mid u \rangle$ over $x \in C$ — the signed distance (scaled by $\|u\|$) you can travel into C along direction u.
- **Polar set** $C^\odot$: the set of directions u for which this maximum never exceeds 1 — a dual object that, remarkably, recovers C itself when you take the polar twice (the *bipolar theorem*, Theorem 7.18).

Outline

Definition 7.1: Support Points and Supporting Hyperplanes I

Definition 7.1

Let C be a nonempty subset of $\mathcal{H}$, let $x \in C$, and suppose that $u \in \mathcal{H} \smallsetminus \{0\}$. If

$$\sup\langle C \mid u\rangle \leq \langle x \mid u\rangle, \tag{7.1}$$

then $\{y \in \mathcal{H} : \langle y \mid u\rangle = \langle x \mid u\rangle\}$ is a **supporting hyperplane** of C at x, and x is a **support point** of C with **normal vector** u. The set of support points of C is denoted $\operatorname{spts} C$, and the closure of $\operatorname{spts} C$ by $\overline{\operatorname{spts}}\, C$.

Plain-English Intuition

Inequality (7.1) says: every point of C scores at most as high as x itself when you measure "progress in direction u" by the inner product $\langle \cdot \mid u\rangle$. Geometrically, the hyperplane through x perpendicular to u is a flat wall that touches C at (at least) the point x, with the whole set C lying on one side. A **support point** is exactly a point where such a wall can be placed — x is "exposed" to the outside along direction u.

Definition 7.1: Support Points and Supporting Hyperplanes II

Proposition 7.2

Let C and D be nonempty subsets of $\mathcal{H}$ such that $C \subset D$. Then $C \cap \operatorname{spts} D \subset \operatorname{spts} C = C \cap \operatorname{spts} \overline{C}$.

Intuition

If x supports the bigger set D, it certainly supports the smaller set $C \subset D$ (the same wall still keeps all of C on one side, since $C \subset D$). And a point of C supports C exactly when it supports the closure $\overline{C}$ — taking closure does not create or destroy support points already inside C.

Proposition 7.3 (Support Points via the Normal Cone)

Let C be a nonempty convex subset of $\mathcal{H}$. Then

$$\operatorname{spts} C = \{x \in C : N_C x \smallsetminus \{0\} \neq \varnothing\} = N_C^{-1}(\mathcal{H} \smallsetminus \{0\}). \tag{7.2}$$

Definition 7.1: Support Points and Supporting Hyperplanes III

Intuition

Recall $N_C x$ collects the directions u for which the supporting inequality (7.1) holds at x (Chapter 6). Proposition 7.3 simply restates Definition 7.1 in that language: x is a support point exactly when its normal cone contains *some* nonzero vector, i.e., when x admits at least one outward-pointing normal direction along which a supporting hyperplane can be erected.

Theorem 7.4 (Bishop–Phelps) and Consequences I

Theorem 7.4 (Bishop–Phelps)

Let C be a nonempty closed convex subset of $\mathcal{H}$. Then

$$\operatorname{spts} C = P_C(\mathcal{H} \smallsetminus C) \quad \text{and} \quad \overline{\operatorname{spts}}\, C = \operatorname{bdry} C. \tag{7.3}$$

Plain-English Intuition

The first identity says: **every** support point of C arises as the metric projection of *some* outside point onto C, and *every* projection of an outside point is a support point (with normal vector pointing from x back toward the source point, by the variational characterization of P_C). The second identity is the headline result: although a boundary point of C need not itself be a support point (Example 7.7 below shows this can fail), the support points are always **dense** in the boundary — you can approximate any boundary point arbitrarily well by genuine support points.

Theorem 7.4 (Bishop–Phelps) and Consequences II

Proposition 7.5

Let C be a convex subset of $\mathcal{H}$ such that $\operatorname{int} C \neq \varnothing$. Then $\operatorname{bdry} C \subset \operatorname{spts} \overline{C}$ and $C \cap \operatorname{bdry} C \subset \operatorname{spts} C$.

Corollary 7.6

Let C be a nonempty closed convex subset of $\mathcal{H}$ and suppose that one of the following holds:
(i) $\operatorname{int} C \neq \varnothing$; (ii) C is a closed affine subspace; (iii) $\mathcal{H}$ is finite-dimensional. Then $\operatorname{spts} C = \operatorname{bdry} C$.

Why This Matters

Corollary 7.6(iii) is the case you will use most in practice: **in finite dimensions, every closed convex set has exactly the support points you would naively expect —** all of $\operatorname{bdry} C$, no more, no less. The subtleties in Theorem 7.4 and Example 7.7 (next slide) are genuinely infinite-dimensional phenomena.

Example 7.7: A Boundary Point That Is Not a Support Point

Example 7.7

Let $\mathcal{H} = \ell^2(\mathbb{N})$ and $C = \ell^2_+(\mathbb{N})$, a nonempty closed convex cone with *empty interior*. For $x = (\xi_k)_{k\in\mathbb{N}} \in \mathcal{H}$, the projection is $P_C x = (\max\{\xi_k, 0\})_{k\in\mathbb{N}}$. By Theorem 7.4, $x = (\xi_k)_{k\in\mathbb{N}} \in C$ is a support point of C if and only if $\{k \in \mathbb{N} : \xi_k = 0\} \neq \varnothing$ (i.e., x has *at least one* zero coordinate). Therefore

$$\left(\frac{1}{2^k}\right)_{k\in\mathbb{N}} \in \operatorname{bdry} C \smallsetminus \operatorname{spts} C :$$

this sequence has every coordinate strictly positive, so it lies on the boundary of the cone (it is not an interior point, since $\operatorname{int} C = \varnothing$ here) but it is *not* a support point, because no coordinate is exactly zero.

Takeaway

This is exactly the phenomenon Corollary 7.6 rules out in finite dimensions: with $\operatorname{int} C = \varnothing$ and $\mathcal{H}$ infinite-dimensional, none of conditions (i)–(iii) of Corollary 7.6 hold, and indeed $\operatorname{spts} C \subsetneq \operatorname{bdry} C$ strictly. **A boundary point of a closed convex set need not be a support point**, even when C is a closed convex cone.

Outline

Definition 7.8: The Support Function

Definition 7.8

Let C be a subset of $\mathcal{H}$. The **support function** of C is

$$\sigma_C : \mathcal{H} \to [-\infty, +\infty] : u \mapsto \sup\langle C \mid u\rangle. \tag{7.4}$$

Plain-English Intuition

Fix a direction u. Among all points $x \in C$, look at the "score" $\langle x \mid u\rangle$ — how far x projects along u (scaled by $\|u\|$). The support function $\sigma_C(u)$ records the *best possible score*: how far into direction u the set C reaches. If C is empty, $\sigma_C \equiv -\infty$ by convention (the sup of the empty set). If C is unbounded in direction u, $\sigma_C(u) = +\infty$. Otherwise $\sigma_C(u)$ is a genuine finite real number for that u.

Why This Function Fully Characterizes Closed Convex Sets

Knowing $\sigma_C(u)$ for *every* $u \in \mathcal{H}$ tells you, for every direction, the exact half-space $\{x : \langle x \mid u\rangle \leq \sigma_C(u)\}$ that must contain C. Intersecting all of these half-spaces recovers C exactly when C is closed and convex (Corollary 7.12) — so σ_C is a complete, direction-indexed "fingerprint" of C.

Example 7.9: Support Function of an Interval in $\mathbb{R}$

Example 7.9

Let Ω be a nonempty subset of $\mathbb{R}$, set $\underline{\omega} = \inf \Omega$ and $\overline{\omega} = \sup \Omega$. Then

$$(\forall \xi \in \mathbb{R}) \quad \sigma_\Omega(\xi) = \begin{cases} \underline{\omega}\, \xi, & \text{if } \xi < 0; \\ 0, & \text{if } \xi = 0; \\ \overline{\omega}\, \xi, & \text{if } \xi > 0. \end{cases} \tag{7.5}$$

Intuition

In $\mathbb{R}$, "direction $u = \xi$" just means "positive" ($\xi > 0$) or "negative" ($\xi < 0$). Walking in the positive direction, the farthest you can score is $\overline{\omega}$ (the supremum of Ω) times the scale ξ; walking in the negative direction, the farthest score comes from the *infimum* $\underline{\omega}$, again scaled by ξ (note $\xi < 0$, so a very negative $\underline{\omega}$ can still give a large positive product). At $\xi = 0$ every point scores 0, so the sup is 0. For instance, if $\Omega = [-2, 5]$, then $\sigma_\Omega(3) = 5 \cdot 3 = 15$ and $\sigma_\Omega(-3) = (-2)(-3) = 6$.

Step-by-Step: The Support Function of the Unit Disk

Example 7.10, Specialized: C = Closed Unit Disk in $\mathbb{R}^2$

Let $C_{\text{disk}} = \{x \in \mathbb{R}^2 : \|x\| \le 1\}$. We claim $\sigma_{C_{\text{disk}}}(u) = \|u\|$ for every $u \in \mathbb{R}^2$: the support function of the unit ball of a norm is the **dual norm**, and the Euclidean norm is (famously) self-dual.

Derivation, by Cauchy–Schwarz

For any x with $\|x\| \le 1$ and any $u \in \mathbb{R}^2$,

$$\langle x \mid u \rangle \le \|x\| \, \|u\| \le \|u\|,$$

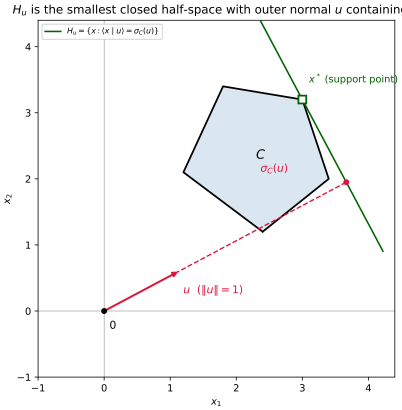
with equality throughout when $x = u/\|u\|$ (a unit vector, so it lies in C_{disk}, and it is aligned exactly with u). Hence the supremum over $x \in C_{\text{disk}}$ is attained and equals $\|u\|$:

$$\sigma_{C_{\text{disk}}}(u) = \sup_{\|x\| \le 1} \langle x \mid u \rangle = \|u\|.$$

Concrete Numbers

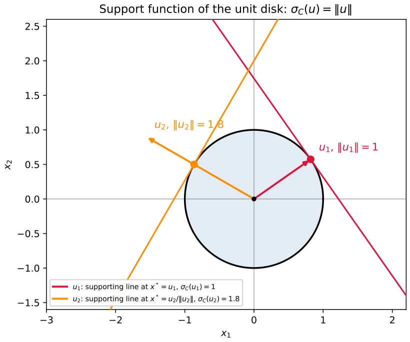
Take $u = (3, 4)$, so $\|u\| = \sqrt{3^2 + 4^2} = 5$. The maximizer is $x^* = u/\|u\| = (0.6,\, 0.8) \in C_{\text{disk}}$, and

Geometric Meaning: The Supporting Half-Space H_u



For $\|u\| = 1$, $H_u = \{x : \langle x \mid u \rangle \leq \sigma_C(u)\}$ is the *smallest* closed half-space with outer normal u that contains C, and $\sigma_C(u)$ is exactly the distance from 0 to the hyperplane $\{x : \langle x \mid u \rangle = \sigma_C(u)\}$ (this is Fig. 7.1 of the book).

The Unit Disk, Revisited: Two Supporting Lines



For u_1 with $\|u_1\| = 1$, the maximizer is $x^* = u_1$ itself and $\sigma_{C_{\text{disk}}}(u_1) = 1$. For u_2 with $\|u_2\| = 1.8$, the maximizer is $x^* = u_2/\|u_2\|$ (still on the unit circle) and $\sigma_{C_{\text{disk}}}(u_2) = 1.8$: the supporting line always sits at distance $\|u\|$ from the origin, perpendicular to u.

Proposition 7.11 and Corollary 7.12: Half-Spaces Reconstruct C I

Proposition 7.11

Let C be a subset of $\mathcal{H}$ and set

$$(\forall u \in \mathcal{H}) \quad H_u = \{x \in \mathcal{H} : \langle x \mid u \rangle \leq \sigma_C(u)\}. \tag{7.6}$$

Then $\overline{\operatorname{conv}}\, C = \bigcap_{u \in \mathcal{H}} H_u$.

Intuition

Each H_u is a closed half-space forced to contain C (that is exactly what $\sigma_C(u)$ measures). Intersecting *all* of them over every possible direction u squeezes down as tightly as possible around C from every side at once — and Proposition 7.11 says this intersection lands *exactly* on the closed convex hull of C, no tighter and no looser.

Proposition 7.11 and Corollary 7.12: Half-Spaces Reconstruct C II

Corollary 7.12

Let C be a closed convex subset of $\mathcal{H}$. Then C is the intersection of all the closed half-spaces of $\mathcal{H}$ that contain C.

Why This Matters

If C is already closed and convex, $\overline{\mathrm{conv}}\, C = C$, so Proposition 7.11 collapses to Corollary 7.12: **a closed convex set is completely determined by the collection of half-spaces containing it**, i.e., by its support function alone. This is the formal statement behind the "fingerprint" claim on the Definition 7.8 slide.

Proposition 7.13: Support Function Is Blind to Taking Convex Hulls

Proposition 7.13

Let C be a subset of $\mathcal{H}$. Then

$$\sigma_C = \sigma_{\operatorname{conv} C} = \sigma_{\overline{\operatorname{conv}}\, C}.$$

Intuition

Taking convex combinations of points in C can never beat the best individual point of C when scoring $\langle \cdot \mid u \rangle$ (a weighted average of scores is at most the largest score being averaged), so $\sigma_{\operatorname{conv} C} \leq \sigma_C$; and clearly $\sigma_C \leq \sigma_{\operatorname{conv} C}$ since $C \subset \operatorname{conv} C$. Passing to the closure adds only limit points, whose scores are themselves limits of scores already achieved in $\operatorname{conv} C$, so the supremum does not change either. In short: **the support function only sees the "outer shape"** of C, and is indifferent to filling in interior points or boundary limit points.

Outline

Definition 7.14: The Polar Set

Definition 7.14

Let C be a subset of $\mathcal{H}$. The **polar set** of C is

$$C^{\odot} = \operatorname{lev}_{\leq 1} \sigma_C = \{u \in \mathcal{H} : (\forall x \in C)\ \langle x \mid u \rangle \leq 1\}. \tag{7.7}$$

Plain-English Intuition

$C^{\odot}$ collects exactly the directions u for which the support function $\sigma_C(u)$ does not exceed 1. Equivalently: $u \in C^{\odot}$ means the half-space $\{x : \langle x \mid u \rangle \leq 1\}$ — the "unit" half-space in direction u — is wide enough to still contain all of C. **The polar set is a dual, direction-indexed shadow of** C: large C (reaching far in many directions) tends to force $C^{\odot}$ to be small (few directions with $\sigma_C(u) \leq 1$), and vice versa.

Consistency Check on C_{disk}

Since $\sigma_{C_{\text{disk}}}(u) = \|u\|$, we get $C_{\text{disk}}^{\odot} = \{u : \|u\| \leq 1\} = C_{\text{disk}}$: the closed unit disk is **self-polar** (Exercise 7.7 in the book).

Example 7.15: The Polar of a Cone Is Its Polar Cone

Example 7.15

Let K be a cone in $\mathcal{H}$. Then $K^{\odot} = K^{\ominus}$ (the polar cone from Chapter 6). In particular, if K is a linear subspace, then $K^{\odot} = K^{\ominus} = K^{\perp}$.

Intuition

For a cone K, if some $x \in K$ scored $\langle x \mid u \rangle > 0$ for a candidate normal u, then scaling x up (still inside K, since K is a cone) would make $\langle x \mid u \rangle$ arbitrarily large, violating the "≤ 1" bound in (7.7). So membership in $C^{\odot}$ for a cone collapses to the strictly stronger condition $\sup\langle K \mid u \rangle \leq 0$, which is exactly the defining condition of the polar cone $K^{\ominus}$ from Chapter 6. For a subspace V, nonnegativity in both directions forces equality, i.e. orthogonality, recovering the familiar $V^{\ominus} = V^{\perp}$.

Running Numeric Example: The Square and Its Polar Diamond

Setup

Let $C_{\mathrm{sq}} = [-1, 1]^2$, the closed unit ball of the sup-norm $\|\cdot\|_\infty$. By Example 7.10 (dual norms), its support function is the *dual* norm of $\|\cdot\|_\infty$, namely the ℓ^1 norm:

$$\sigma_{C_{\mathrm{sq}}}(u) = |u_1| + |u_2| = \|u\|_1.$$

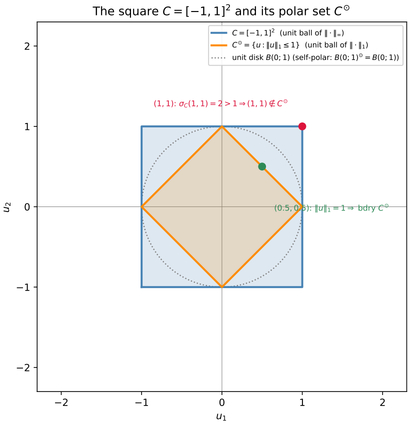
Two Numeric Checks of $\sigma_{C_{\mathrm{sq}}}$

For $u = (1, 1)$: $\sigma_{C_{\mathrm{sq}}}(1, 1) = \sup_{x \in C_{\mathrm{sq}}}(x_1 + x_2) = 1 + 1 = 2$, attained at $x^* = (1, 1)$; and indeed $\|(1, 1)\|_1 = 2$. For $u = (2, -1)$: the maximizer is $x^* = (1, -1)$, giving $\sigma_{C_{\mathrm{sq}}}(2, -1) = 2 \cdot 1 + (-1)(-1) = 3 = \|(2, -1)\|_1$.

The Polar Set: From $\sigma_{C_{\mathrm{sq}}}$ to $C_{\mathrm{sq}}^{\odot}$

By (7.7), $C_{\mathrm{sq}}^{\odot} = \{u : \|u\|_1 \le 1\}$, the closed unit ball of $\|\cdot\|_1$ — a diamond with vertices $(\pm 1, 0)$, $(0, \pm 1)$. Check $u = (1, 1)$: $\|(1, 1)\|_1 = 2 > 1$, so $(1, 1) \notin C_{\mathrm{sq}}^{\odot}$ (consistent with $\sigma_{C_{\mathrm{sq}}}(1, 1) = 2 > 1$). Check $u = (0.5, 0.5)$: $\|(0.5, 0.5)\|_1 = 1$, so this point sits exactly on $\operatorname{bdry} C_{\mathrm{sq}}^{\odot}$.

The Square and Its Polar, Pictured



The square $C_{\mathrm{sq}} = [-1, 1]^2$ (blue) and its polar $C_{\mathrm{sq}}^\odot = \{u : \|u\|_1 \le 1\}$ (orange), with the unit disk (dotted) shown for reference as the self-polar case $B(0;1)^\odot = B(0;1)$.

Proposition 7.16: Basic Properties of the Polar I

Proposition 7.16

Let C and D be subsets of $\mathcal{H}$. Then:

(i) $C^{\perp} \subset C^{\ominus} \subset C^{\odot}$.

(ii) $0 \in C^{\odot}$, and $C^{\odot}$ is closed and convex.

(iii) $C \cup \{0\} \subset C^{\odot\odot}$.

(iv) If $C \subset D$, then $D^{\odot} \subset C^{\odot}$ and $C^{\odot\odot} \subset D^{\odot\odot}$.

(v) $C^{\odot\odot\odot\odot} = C^{\odot}$.

(vi) $(\overline{\operatorname{conv}}\, C)^{\odot} = C^{\odot}$.

Proposition 7.16: Basic Properties of the Polar II

Intuition, Item by Item

(ii): $u = 0$ trivially satisfies $\langle x \mid 0 \rangle = 0 \leq 1$ for every x, and $C^{\odot}$ is an intersection of closed half-spaces (one per $x \in C$), hence closed and convex automatically. (iii): applying the defining inequality of $C^{\odot\odot}$ to elements of $C^{\odot}$ shows every point of C (and 0) satisfies it. (iv): **polarity reverses inclusions** — a bigger set imposes more constraints, shrinking its polar; this is the same "duality flips order" pattern seen for polar cones in Chapter 6. (vi): matches Proposition 7.13 — the polar, like the support function it is built from, cannot distinguish C from its closed convex hull.

Proposition 7.16: Basic Properties of the Polar III

Remark 7.17 (Specializing to Cones)

Combining Example 7.15 with Proposition 7.16, for an arbitrary cone K: $K \subset K^{\ominus\ominus}$, $K^{\ominus\ominus\ominus} = K^{\ominus}$, and $(\operatorname{conv} K)^{\ominus} = K^{\ominus}$. For a linear subspace V: $V \subset V^{\perp\perp}$, $V^{\perp\perp\perp} = V^{\perp}$, and $\overline{V}^{\perp} = V^{\perp}$.

Theorem 7.18 (The Bipolar Theorem) and Its Corollary

Theorem 7.18 (Bipolar Theorem)

Let C be a subset of $\mathcal{H}$. Then

$$C^{\odot\odot} = \overline{\operatorname{conv}}\,(C \cup \{0\}).$$

Plain-English Intuition

Polarizing twice does not quite return C in general — it returns the *smallest closed convex set containing* C *and the origin*. Any "non-convexity," "non-closedness," or "missing the origin" in C gets repaired by the double-polar operation, but nothing beyond that changes. This is the exact convex-analytic analogue of the biorthogonal-complement fact $V^{\perp\perp} = \overline{V}$ for subspaces.

Corollary 7.19

Let C be a subset of $\mathcal{H}$. Then: (i) C is closed, convex, and contains the origin $\iff C^{\odot\odot} = C$; (ii) C is a nonempty closed convex cone $\iff C^{\ominus\ominus} = C$; (iii) C is a closed linear subspace $\iff C^{\perp\perp} = C$.

Bipolar Theorem, Checked on the Running Examples

$C_{\text{sq}} = [-1,1]^2$

C_{sq} is already closed, convex, and contains 0, so Corollary 7.19(i) predicts $C_{\text{sq}}^{\odot\odot} = C_{\text{sq}}$. Indeed, we computed $C_{\text{sq}}^{\odot} = \{u : \|u\|_1 \le 1\}$ (the ℓ^1 diamond); polarizing once more returns the sup-norm ball, since the dual of $\|\cdot\|_1$ is $\|\cdot\|_\infty$: $C_{\text{sq}}^{\odot\odot} = \{x : \|x\|_\infty \le 1\} = C_{\text{sq}}$.

$C_{\text{disk}} = B(0;1)$

Self-polar, so trivially $C_{\text{disk}}^{\odot\odot} = C_{\text{disk}}^{\odot} = C_{\text{disk}}$, again consistent with Corollary 7.19(i).

A Non-Convex Sanity Check

Let $C = \{(1,0),(0,1)\}$ (two points, not convex, not containing 0). Then $\sigma_C(u) = \max\{u_1, u_2\}$, so $C^{\odot} = \{u : u_1 \le 1 \text{ and } u_2 \le 1\}$, and one finds $C^{\odot\odot} = \overline{\text{conv}}(C \cup \{0\})$: the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$ — exactly Theorem 7.18's prediction, and visibly larger than C itself.

Outline

Python: Support Functions and Polar Sets, Numerically

The code below computes $\sigma_C(u)$ by brute-force maximization over a fine sample of each set's boundary, verifies the disk formula $\sigma_{C_{\mathrm{disk}}}(u) = \|u\|$ and the square formula $\sigma_{C_{\mathrm{sq}}}(u) = \|u\|_1$, and checks numerically that $C_{\mathrm{sq}}^{\odot\odot} = C_{\mathrm{sq}}$ (Corollary 7.19(i)) by sampling many directions.

```
import numpy as np
rng = np.random.default_rng(0)

# --- sigma_C via brute-force sampling of C's boundary/interior -----------
def sigma_disk(u):                             # closed unit disk
    return np.linalg.norm(u)                   # exact: sigma(u) = ||u||

def sigma_square(u):                           # C_sq = [-1,1]^2
    return abs(u[0]) + abs(u[1])               # exact: sigma(u) = ||u||_1

# check against brute-force sup over many sampled points of each set
disk_pts = np.array([[np.cos(t), np.sin(t)]
                     for t in np.linspace(0, 2*np.pi, 4000)])
sq_pts = rng.uniform(-1, 1, size=(20000, 2))
sq_pts = np.vstack([sq_pts, [[1,1],[1,-1],[-1,1],[-1,-1]]])  # incl. corners

max_err_disk, max_err_sq = 0.0, 0.0
for _ in range(200):
    u = rng.uniform(-3, 3, size=2)
    bf_disk = np.max(disk_pts @ u)
    bf_sq = np.max(sq_pts @ u)
    max_err_disk = max(max_err_disk, abs(bf_disk - sigma_disk(u)))
    max_err_sq = max(max_err_sq, abs(bf_sq - sigma_square(u)))
```

Outline

The Three Big Objects

Object	Role
$\operatorname{spts} C$	support points; dense in $\operatorname{bdry} C$ always (Bishop–Phelps), equal to $\operatorname{bdry} C$ in finite dimensions
$\sigma_C(u) = \sup\langle C \mid u\rangle$	support function; a complete, direction-indexed fingerprint of $\overline{\operatorname{conv}}\, C$
$C^{\odot} = \{u : \sup\langle C \mid u\rangle \leq 1\}$	polar set; dual shadow of C, closed and convex by construction

Chapter Summary II

The Key Identities

- $\overline{\operatorname{conv}}\, C = \bigcap_u \{x : \langle x \mid u \rangle \leq \sigma_C(u)\}$ (Proposition 7.11): half-spaces reconstruct the closed convex hull.
- $\sigma_C = \sigma_{\operatorname{conv} C} = \sigma_{\overline{\operatorname{conv}}\, C}$ (Proposition 7.13): the support function is blind to convexifying or closing C.
- $C^{\odot\odot} = \overline{\operatorname{conv}}\,(C \cup \{0\})$ (Theorem 7.18, the bipolar theorem): double polarity repairs non-convexity, non-closedness, and a missing origin — nothing else.
- For a cone K: $K^{\odot} = K^{\ominus}$ (Example 7.15); for a subspace V: $V^{\odot} = V^{\ominus} = V^{\perp}$.

Chapter Summary III

Running Examples, Side by Side

Set C	$\sigma_C(u)$	$C^{\odot}$
$B(0;1)$ (unit disk)	$\|u\|$	$B(0;1)$ (self-polar)
$[-1,1]^2$ (unit square)	$\|u\|_1$	$\{u : \|u\|_1 \leq 1\}$ (diamond)

What Comes Next

Chapter 8 turns from *sets* to *functions*: convex functions $f : \mathcal{H} \to [-\infty, +\infty]$ are introduced via their epigraphs, and the support function σ_C of this chapter will reappear as a first, prototypical example of a convex function built from a set — a preview of the **conjugate function** machinery of Chapters 13–15.